

nanochat Precision Racecar



**A B300-ready product for controlled
BF16 vs FP8 vs FP4 pretraining.**

Prestage assets, fail fast on Blackwell, train multi-node, and emit comparison artifacts a judge can read in one minute.

FLOW
prestage

FLOW
smoke

FLOW
compare

Judge-facing output

```
BASE_DIR=/shared/nanochat/b300 \  
RUN_NAME=track1-demo \  
bash runs/fog_compare_b300.sh
```

Artifacts:

```
comparisons/<run>/comparison.json  
comparisons/<run>/comparison.md  
base_checkpoints/<model_tag>/
```

What Judges Actually Score

Strong hackathon demos look like products, not terminal archaeology

- A clean runbook another team can execute without your verbal explanation.
- A smoke test that fails on the real reasons teams lose time: wrong GPU class, missing FA4, bad rendezvous, unsupported precision path.
- One artifact that makes the result legible: same tokenizer, same shard order, same seed, same batch schedule, only precision changes.
- A crisp claim: better throughput under controlled settings, with quality drift visible instead of hand-waved.

The one-line thesis

Track 1 becomes a controlled product: same pipeline, same data, same seed, only precision changes.

Judge shortcuts we optimize for

- Can it start cleanly?
- Can I see the comparison in one file?

Product Surface

1. Prestage shared assets

```
runs/b300_prestage.sh
```

Checks Blackwell runtime, pulls dataset shards, trains tokenizer, and leaves a clean BASE_DIR for the timed window.

2. Fail fast before scoring

```
runs/b300_smoke.sh
```

Exercises the intended FP8 stack on one node so the real cluster slot is not spent discovering missing FA4 or FP8 incompatibilities.

3. Train and compare

```
runs/b300_train.sh  
runs/b300_eval.sh  
runs/fog_compare_b300.sh
```

Produces checkpoints plus comparison.json and comparison.md for judge-ready inspection.

Controlled Precision Lab

Fixed across all three arms

- Tokenizer and dataset shards
- Random seed and shard order
- Architecture family and depth
- Total batch size and token budget
- Optimizer schedule and eval cadence
- Comparison metrics: tok/sec, step time, wall time, val_bpb

Only one variable moves

RECIPE

bf16

RECIPE

fp8

RECIPE

fp4

The deck claim is simple: when a judge asks whether FP8 or FP4 actually helped, we can

Artifact schema

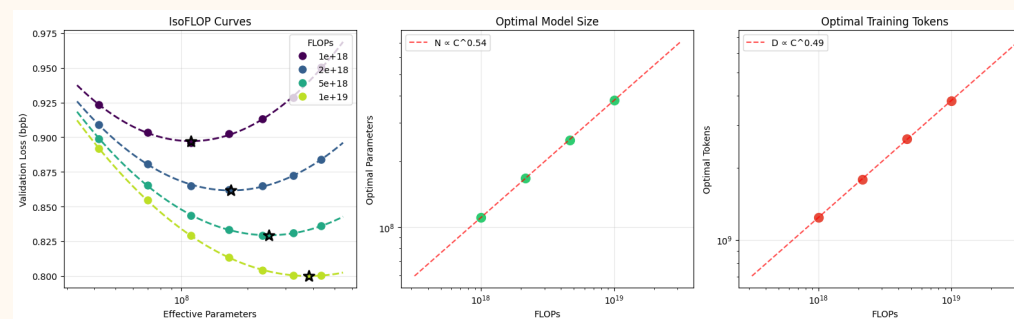
```
{
  "recipe": "fp8_full",
  "tok_per_sec": 0.0,
  "step_time_s": 0.0,
  "val_bpb": 0.0
}
```

Why This Is Credible

This is not a slideware rewrite.

- The fork already carries B300 prestige, smoke, train, and eval entrypoints.
- The FOG family exists separately from default nanochat, so the precision experiment does not get polluted by unrelated architecture tricks.
- The comparison harness already writes machine-readable artifacts for side-by-side throughput and quality.
- The upstream nanochat story remains intact, which makes the fork legible to other contributors.

Upstream credibility + Blackwell extension



nanochat already has a speedrun identity. The product pitch is that this fork extends that identity to Blackwell-era

BASE
nanochat

FORK
B300

STORY
product

Live Demo In 60 Seconds

1. Bring the node up cleanly

```
BASE_DIR=/shared/nanochat/b300 \  
bash runs/b300_smoke.sh
```

```
# proves:  
# - Blackwell detected  
# - FlashAttention-4 active  
# - precision stack usable
```

2. Show the product artifact

```
cat $BASE_DIR/comparisons/track1/comparison.md
```

recipe	tok/sec	step_s	val_bpb
bf16	...	...	...
fp8	...	...	...
fp4	...	...	...

The visual proof is compact: one command to validate the system, one file to understand the result.

Why This Can Win

Most teams will present a run.

We can present a reusable training product.

- Judges can see the operational path from prestige to comparison without reverse-engineering our terminals.
- Sponsors can imagine this becoming a public Blackwell benchmark, not just a hackathon artifact.
- The prize hardware extends the same story cleanly: larger matched sweeps, stronger evidence, public reproducibility.

Repo

EvergreenTree/nanochat

Pitch line

A Blackwell-ready control tower for fair low-precision pretraining comparisons.

Ask

Back the team shipping product, not just a run.